Reproducibility: Heart Failure Condition and Survival Analysis

Group 3 - Data 6550 - Spring 2026
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Statement on AI Use: AI such as ChatGPT was used in this project, such as for assisting with comprehension of some of the modeling methodologies, cleaning up prose and expanding on insights.

1. Introduction

a. Paper and Data Summary

The paper that we based our reproducibility project on is *A Comparative Study for Time-to-Event Analysis and Survival Prediction for Heart Failure Conditions using Machine Learning Techniques* (Mishra, 2022a). As the title suggests, the paper's objective is to "build a survival analysis system and survival classification system for heart failure that helps in the timely and accurate prediction of the survival function." To do so, the researchers implement techniques like Kaplan-Meier analysis and Cox Potential Hazard regression models to estimate the time to death event for heart failure, as well as several classification models (SVM, decision tree, random forest, XGBoost, and LightGBM) to predict the survival of patients. The purpose of the study is to "aid the physicians and act as a decision support system for making better assessments and thereby increasing the patient outcome levels."

The dataset used for the study (Ahmad, et al, 2017) contains cardiovascular medical records taken from 299 patients with a history of class III or class IV heart failure. The records contain 105 women and 194 men between the ages of 40 and 95. There are 13 features in the dataset, which are: Age, Anemia, High Blood Pressure, Creatinine phosphokinase, Diabetes, Ejection Fraction, Sex, Platelets, Serum Creatinine, Serum Sodium, Smoking, Time, and Death Event. A more thorough description of the features can be found in the appendix at the end of this paper.

b. Paper Results Summary

A series of experiments were conducted to analyze the survival probability using the Kaplan-Meier (KM) estimates and Cox regression models. The KM estimate survival curve is defined as "the probability of surviving in a given length of time while considering time in many small intervals". It may question, for example, "Let's compare survival between smokers vs non-smokers". Cox Proportional Hazards Model "allows to compute survival regressions on the time duration using the censored subjects and the features available as part of the dataset based on small intervals containing at most one event of interest." The Cox model would pose the earlier question in the context, "Let's estimate the effect of smoking *while controlling for age, health, etc.*"

KM Results: The paper concludes from the KM modeling that Anemia, High Blood Pressure, Ejection Fraction, Serum Creatinine, and Time are known to have a major impact on the survival probability of patients who are at an advanced stage of heart failure. Figures 1 & 2 (in the appendix section) illustrates these findings.

Cox Model Results: The results from the Cox models show that age, serum creatinine, and ejection fraction (EF) are the most significant predictors of mortality in heart failure patients.

- Age increases death risk by ~3.9% per year
- Serum creatinine increases risk by ~28.1% per unit (strong negative impact)
- Ejection fraction (EF) decreases risk by ~5.2% per unit (protective effect)
- Anaemia is a meaningful risk factor, increasing mortality risk by about 61%.

Other variables—smoking, sex, platelets, and diabetes—were not statistically significant due to high p-values and uncertainty in estimates. Figure 3 (in the appendix section) demonstrates the graphed results of the key variables mentioned.

ML classification results: The several categorical prediction models aimed to predict death events performed as seen in Figure 4. Additionally, feature importance was assessed, and the paper concluded that a majority of the models assign the highest scores to serum creatinine, ejection fraction, and age. Figure 5 shows how the feature importance stacks up in the Light GBM Model

Model	Confusion Matrix	F1 Score	Accuracy	Precision	Recall	MCC	AUC ROC	AUC PR
SVM	0 6 3 1 2 19 0 1	88.37	83.33	86.36	90.48	59.22	78.6	74.2
Decision Tree	0 5 4 1 5 16 0 1	78.05	70.00	80.00	76.19	30.86	65.9	86.4
Random Forest	0 7 2 1 4 17 0 1	85.00	80.00	89.47	80.95	55.85	79.4	83.8
XGBoost	0 7 2 1 4 17 0 1	85.00	80.00	89.47	80.95	55.85	79.4	83.8
Light GBM	0 6 3 1 3 16 0 1	85.71	80.00	85.71	85.71	52.38	76.2	80.14

Figure 4: ML Classification Results

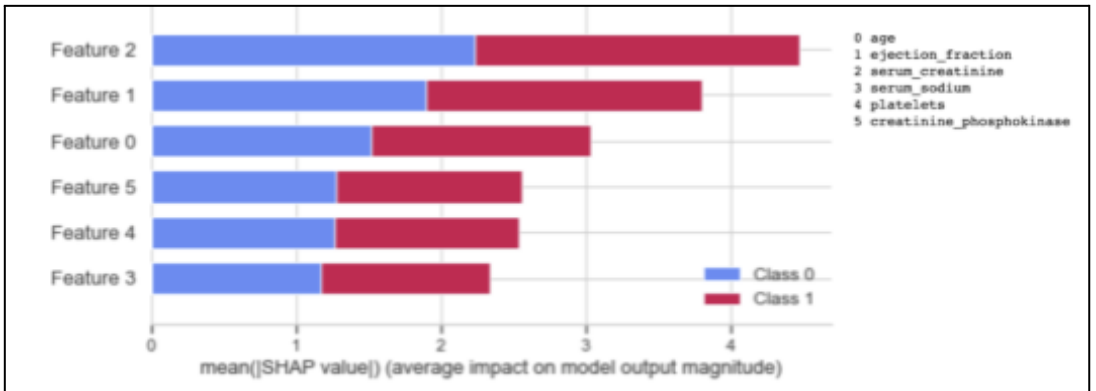


Figure 5: Impact of Features on Output

c. Reproduction goals

Our goal in this project is to try to run the models described above (KM, Cox and Various ML algorithms) and determine if our outputs successfully match (or identify to what degree they failed to match).

2. Methodology

a. Reproduction Approach

The author of the paper provides strong intel that should aid with reasonably successful reproducibility, starting out with spelling out the system design and architecture. The paper provides specificities of the environment, clarifying the CPUs, GPUs, and RAM of the cloud-based system; the version of Python used (3.8.8) and the miscellaneous libraries used. The paper also specifies the

machine learning algorithms and although it doesn't inform of every tuning parameter, it informs us that the "modes are tuned to extract the best learning parameters and the best estimator using the RandomizedSearchCV approach where not all hyperparameter values are considered. Instead, a set number of hyperparameter settings is experimented from the specified probability distributions. A 5-fold cross-validation configuration is used to fit the RandomizedSearchCV method."

Additionally, the author also published a GitHub repository containing thorough code related to the project, which we utilized for this project (Mishra, 2022b). Utilization of this resource was especially helpful in our replication attempt.

b. Environmental Setup

The reproduction of this paper was conducted using Python in Visual Studio Code with Jupyter notebook support. A virtual environment was created for reproducibility and isolation of dependencies. The necessary libraries were installed using pip, depending on the requirements specified in the repository under requirements. txt, and additional libraries not specified in the file but were necessary for execution.

Some of the key libraries used include pandas for data manipulation, numpy for numerical calculations, scikit-learn for building models, and lifelines for Cox regression and Kaplan-Meier analysis. For visualisation, Matplotlib and Seaborn were used, and additional libraries such as imbalance-learn, XGBoost, LightGBM, and SHAP were installed for classification model development and additional feature importance analysis.

Some libraries, such as Kaleido, were installed manually to enable figure export, and all experiments were carried out in the environment to ensure reproducibility.

3. Results

a. Comparison with Original

<u>Kaplan-Meier (KM) Analysis:</u>

The Kaplan-Meier results appear similar to the original study and show the same survival trends. We observed that survival probability decreases over time, which matches the paper. Important factors like serum creatinine, high blood pressure, ejection fraction, and anemia affect survival. Patients with higher creatinine and lower ejection fraction had lower survival, meaning higher death risk. Overall, there were only small differences in numerical insights. This indicates that the relative influence of these variables on survival was successfully reproduced, even if exact survival probabilities differed slightly.

<u>Cox Proportional Hazards</u> :

The Cox regression results are consistent with the original study and show similar trends. We found that age and serum creatinine increase the risk of death, while ejection fraction improves survival. Anemia and high blood pressure also increase mortality risk. Variables like diabetes, sex, and smoking were not statistically significant. Overall, we matched the original study and confirmed the same key predictors of survival. The consistency of statistically significant predictors suggests that the underlying relationships between features and survival outcomes are robust to minor implementation differences.

Machine learning models:

Random Forest and XGBoost showed the strongest performance with accuracy = 0.85 and F1 = 0.7429, while LightGBM with accuracy = 0.8167 and F1 = 0.7027, and SVM had moderate results with accuracy = 0.80 and F1 = 0.6842. Decision Tree performed the lowest with accuracy = 0.7833 and F1 = 0.6486. Feature importance consistently identified time, ejection fraction, serum creatinine, and age as the most important predictors. Overall, our results support the original study and confirm that ensemble models are more effective for this task.

Comparison Table - ML Scores Between the Paper and the Replication

Model	Paper Accuracy	Replication Accuracy	Accuracy Difference	Paper Precision	Replication Precision	Precision Difference	Paper Recall	Replication Recall	Recall Difference
Random Forest	0.8	0.85	0.05	0.8947	0.8125	-0.0822	0.8095	0.6842	-0.1253
XGBoost	0.8	0.85	0.05	0.8947	0.8125	-0.0822	0.8095	0.6842	-0.1253
LightGBM	0.8	0.8167	0.0167	0.8571	0.7222	-0.1349	0.8571	0.6842	-0.1729
SVM	0.8333	0.8	-0.0333	0.8636	0.6842	-0.1794	0.9048	0.6842	-0.2206
Decision Tree	0.7	0.7833	0.0833	0.8	0.6667	-0.1333	0.7619	0.6316	-0.1303

Model	Paper F1	Replication F1	F1 Difference	Paper ROC-AUC	Replication ROC-AUC	ROC-AUC Difference
Random Forest	0.85	0.7429	-0.1071	0.794	0.9076	0.1136
XGBoost	0.85	0.7429	-0.1071	0.794	0.8742	0.0802
LightGBM	0.8571	0.7027	-0.1544	0.762	0.8652	0.1032
SVM	0.8837	0.6842	-0.1995	0.786	0.8896	0.1036
Decision Tree	0.7805	0.6846	-0.0959	0.659	0.8042	0.1452

*Note: difference defined as replication result - Paper result

When we drill down to each model's individual scores, some key observations identified in the machine learning results comparison are:

- Replication results of *accuracy* remained the most similar amongst all parameters with a margin of no more than 0.08.
- *Precision, Recall* and *F1* of the replication was consistently poorer than the paper by approximately a 0.1 to 0.2 margin.
- *ROC-AUC* of the replication showed better performance by approximately a 0.1 margin.

These discrepancies suggest that while model ranking and feature importance are stable, threshold-dependent metrics are more sensitive to implementation details.

b. Discrepancy Analysis

Potential factors contributing to differing results:

1. Random train/test split could yield different results. Especially considering the relatively small observation size of 299 patients, even minor differences in how the data is partitioned can significantly affect model evaluation metrics.
2. Different program versions or environments may have contributed to minor differences in replication of the paper's results. For the purpose of this project, and with the time constraints, our team didn't feel that it was necessary to completely mimic the author's environment.

3. The author didn't specify all tuning parameters for the machine learning models used. While the authors mention using RandomizedSearchCV, the absence of full parameter specifications makes exact replication difficult.
4. With the exception of the machine learning models' results (accuracy, precision, recall, etc.) the paper didn't always specify in detail about the numerical results of the KM and Cox models, but rather clarified what parameters came up as significant. Although upon running the replication, we arrived at similar conclusions, it is not clear if the results were exact, numerically.
5. Additionally, some machine learning models such as Random Forest and XGBoost involve randomness, which can lead to slight variations in results even with the same data.
6. Small differences in preprocessing steps, such as scaling, feature selection, or handling of variables, may also contribute to variations in numerical results. Also on this note, it was noticed that in our replication of the ML prediction models, we failed to remove "time" from the model, which would have been a data leakage variable. It was mentioned in the paper that it was removed, but we overlooked this detail. Because "time" is directly related to the outcome (death event), its inclusion introduces data leakage, artificially improving model performance. This likely explains why our ROC-AUC scores were higher than those reported in the paper.

<u>What helped some things remain consistent:</u>

1. Static Dataset: Exploratory data analysis, essentially descriptive statistics remained consistent. The full and un-changed dataset that was used in the paper was available for use by our team for the replication attempt.
2. Clear methodology: The original paper clearly described key methods such as Kaplan-Meier analysis, Cox regression, and machine learning models, which helped in reproducing the main steps. Also, by providing a Github and sharing code, this contributed greatly to a reasonably successful replication of the study.
3. Strong feature relationship: Important variables like age, serum creatinine, and ejection fraction showed strong relationships with survival, making the overall trends consistent even when numerical values varied slightly.

Despite these discrepancies, the replication confirms that:

- Core clinical predictors (age, serum creatinine, ejection fraction) are consistently identified across methods
- Model rankings (ensemble > single models) remain stable
- Survival analysis methods (KM, Cox) are more reproducible than ML metrics

This suggests that interpretability-focused models (KM, Cox) may be more robust for clinical insights, while machine learning models are more sensitive to implementation details and preprocessing choices.

4. Ethics Discussion:

a. Reproducibility Challenges

Reproducibility is an important component of ethical data science, especially in healthcare contexts where model outputs may influence clinical decision-making. In this study, the ability to reproduce similar trends and conclusions strengthens confidence in the original findings. However, the small

discrepancies observed, as detailed in section 3, highlight an important ethical concern: even minor variations in implementation can lead to different conclusions, which may ultimately impact patient outcomes if such models are deployed in practice.

b. Resource Considerations

The project required computational resources for running machine learning models and performing survival analysis. However, the dataset size was manageable, and tools like Python made the implementation feasible without high-end hardware. The fact that the paper's author published a GitHub repository along with the paper was especially helpful and is a behavior we would like to continue to practice in the field of data science. This allows peers in the scientific community to replicate one's work to reinforce, question, and/or build upon the work, which is a key part of the scientific process.

c. Scientific Integrity Implications

Overall, this replication study reinforces that ethical data science is not only about achieving accurate results, but also about ensuring that those results are transparent, reproducible, and interpretable. In high-stakes domains such as healthcare, even small methodological oversights, such as data leakage or incomplete reporting, can have outsized consequences, underscoring the need for rigorous standards in both research and implementation. Creating projects that are well-documented and replicable is something we should all strive to achieve so that our findings can more completely contribute to scientific knowledge and withstand integrity tests.

5. Recommendations

a. Best Practices

Future studies should prioritize complete transparency in methodology. This includes:

- Providing full hyperparameter configurations.
- Reporting random seeds.
- Clearly documenting preprocessing steps (e.g., scaling, feature removal).
- Explicitly identifying and removing potential data leakage variables.

Additionally, sharing well-documented code repositories significantly improves reproducibility, as demonstrated in this study.

b. Process Improvements

Several improvements could strengthen future research:

- Use larger datasets: Small datasets increase variability and reduce generalizability.
- Standardizing evaluation procedures, using consistent cross-validation instead of single splits would reduce variability.
- Separate survival analysis and classification tasks clearly, avoiding overlap of variables like "time" that can introduce leakage.
- Include confidence intervals and uncertainty estimates for model metrics.

6. References

Ahmad, Tanvir; Munir, Assia; Bhatti, Sajjad Haider; Aftab, Muhammad; Ali Raza, Muhammad (2017): DATA_MINIMAL.. PLOS ONE. Dataset- https://doi.org/10.1371/journal.pone.0181001.s001 Link: https://plos.figshare.com/articles/dataset/Survival_analysis_of_heart_failure_patients_A_case_study/5227684/1

Mishra, S. (2022a) "A Comparative Study for Time-to-Event Analysis and Survival Prediction for Heart Failure Condition using Machine Learning Techniques", Journal of Electronics, Electromedical Engineering, and Medical Informatics, 4(3), pp. 115-134. doi: 10.35882/jeeemi.v4i3.225. https://jeeemi.org/index.php/jeeemi/article/view/225/94

Mishra, S. (2022b). Heart-Failure-Condition-And-Survival-Analysis [Software]. GitHub.https://github.com/sauravmishra1710/Heart-Failure-Condition-And-Survival-Analysis/tree/master.

Appendix

Dataset Features

Feature	Explanation	Measurement	Range
Age	Age of the patient	Years	[40 -- 95]
Anaemia	Decrease of red blood cells or haemoglobin level in the blood.	Binary	0, 1
High Blood Pressure	Indicative of whether the patient has hypertension.	Binary	0, 1
Creatinine Phosphokinase	Level of the CPK enzyme in the blood	mcg/L	[23 -- 7861]
Diabetes	If the patient is diabetic	Binary	0, 1
Ejection Fraction	Percentage of blood leaving the heart at each contraction	Percentage	[14 -- 80]
Sex	Gender of the patient.	Binary	0, 1
Platelets	Platelets in the blood	kiloplatelets/mL	[25.01 -- 850.00]
Serum Creatinine	Level of creatinine in the blood	mg/dL	[0.50 -- 9.40]
Serum Sodium	Level of sodium in the blood	mEq/L	[114 -- 148]
Smoking	Indicates if the patient has smoking habit.	Binary	0, 1
Time	Follow-up period for the next doctor visit.	Days	[4 -- 285]
Death Event	If the patient died during the follow-up period.	Binary	0, 1

Figures:

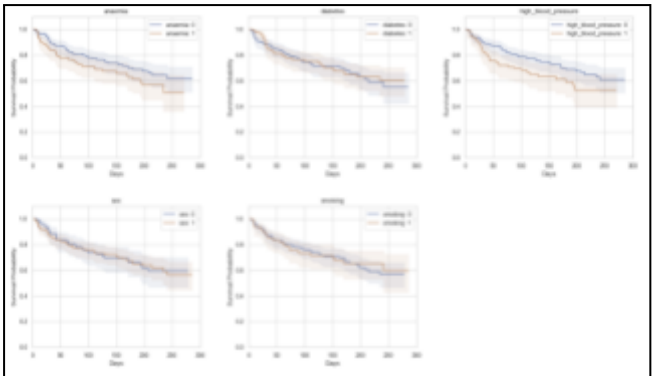


Figure 1: Paper's KM Estimates - Category Features

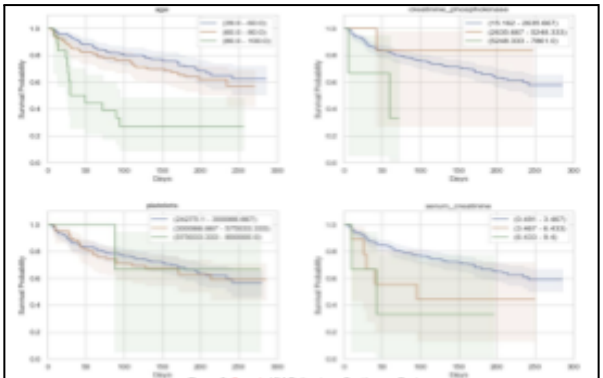


Figure 2: Paper's KM Estimates - Continuous Features

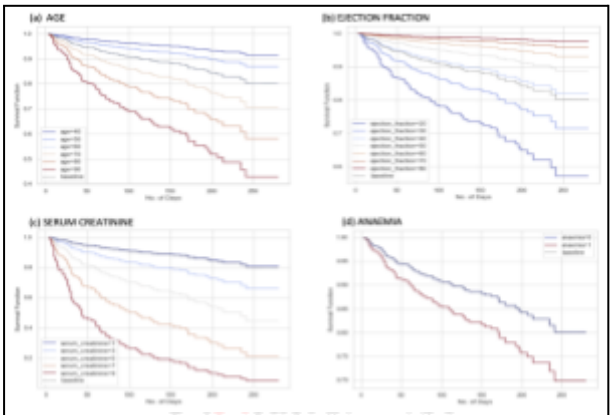
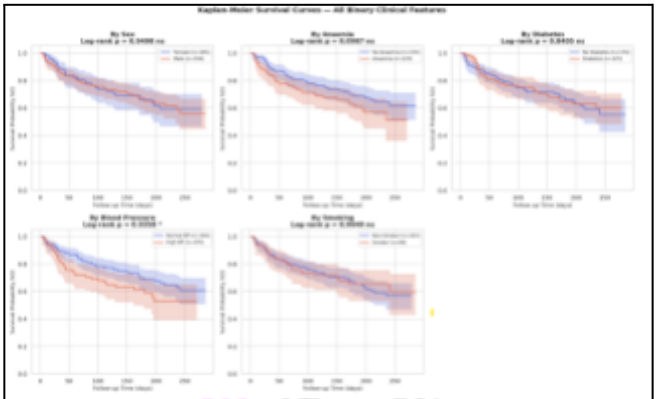


Figure 3: Paper's Partial effects of features on survival function



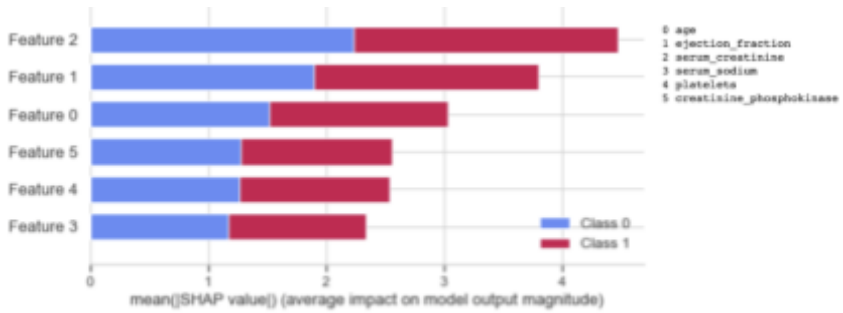
Replication results: KM curves appear identical

```
================================================================
  COX PH REGRESSION — KEY FINDINGS
================================================================
                          HR  HR_lower  HR_upper  p_value  sig
covariate
age                   1.0475    1.0285    1.0668   0.0000  ***
ejection_fraction     0.9522    0.9329    0.9720   0.0000  ***
serum_creatinine      1.3786    1.2014    1.5818   0.0000  ***
creatinine_phosphokinase 1.0002 1.0000    1.0004   0.0260    *
high_blood_pressure   1.6092    1.0534    2.4583   0.0278    *
anaemia               1.5843    1.0358    2.4233   0.0338    *
serum_sodium          0.9568    0.9141    1.0014   0.0575   ns
sex                   0.7886    0.4816    1.2913   0.3452   ns
diabetes              1.1501    0.7427    1.7811   0.5307   ns
smoking               1.1376    0.6953    1.8614   0.6078   ns
platelets             1.0000    1.0000    1.0000   0.6806   ns

C-index (discrimination): 0.7408

Interpretation guidance:
  HR > 1 → Higher hazard / worse survival
  HR < 1 → Lower hazard  / better survival
  HR = 1 → No effect
```

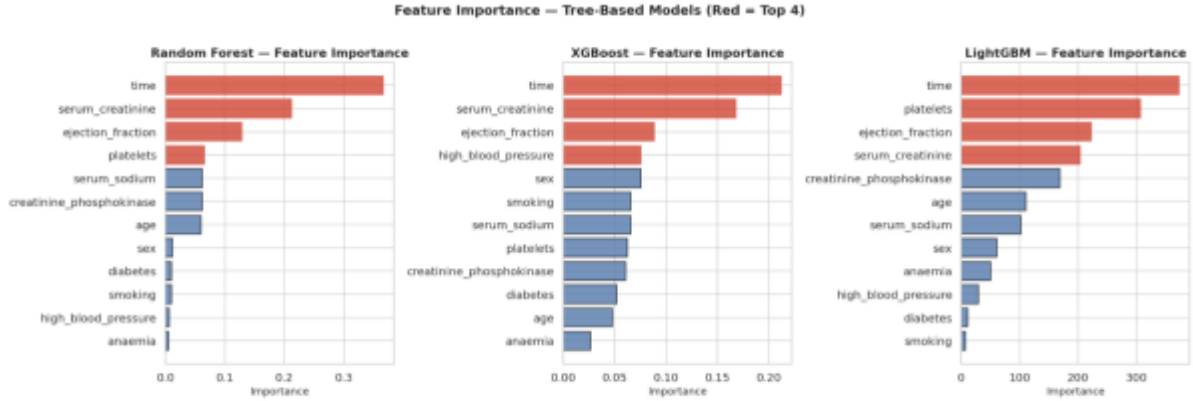
Replication Results: Cox Findings

```
================================================================
  MACHINE LEARNING — FINAL RESULTS SUMMARY (All Features)
================================================================
              Accuracy  Precision  Recall      F1  ROC-AUC     MCC
Model
Random Forest    0.8500     0.8125  0.6842  0.7429   0.9076  0.6428
XGBoost          0.8500     0.8125  0.6842  0.7429   0.8742  0.6428
LightGBM         0.8167     0.7222  0.6842  0.7027   0.8652  0.5707
SVM              0.8000     0.6842  0.6842  0.6842   0.8896  0.5379
Decision Tree    0.7833     0.6667  0.6316  0.6486   0.8042  0.4926

── Best Model: Random Forest ──
Accuracy     0.8500
Precision    0.8125
Recall       0.6842
F1           0.7429
ROC-AUC      0.9076
MCC          0.6428

Key Observations:
────────────────
• All 5 models achieve reasonable performance on this small dataset (n=299)
• Ensemble models (RF, XGBoost, LightGBM) generally outperform single models
• Top features: time, ejection_fraction, serum_creatinine, age
• SMOTE improves recall for the minority class (Deceased)
• More data would improve model stability and robustness
```

Replication Results: ML scores



Paper's Feature Impact of LightGBM Model



Replication Results - Tree-based model feature importance